

↳ maximum product subarray. → any contiguous

arr[] = [2, 3, -2, 4]

ans = 6

↳ subarray

part → whole array can

be a subarray too.

Brute force. $maxi = INT_MIN;$

for (i → n)

{

for (j → i → n)

prod = 1

{

for (k = i = j)

{

prod = prod * arr[k]

}

maxi = max(maxi, prod)

}

}

$T.C = O(n^3)$ → $O(n^2)$

$S.C \rightarrow O(1)$

for (i → n)

{

for (j → i → n)

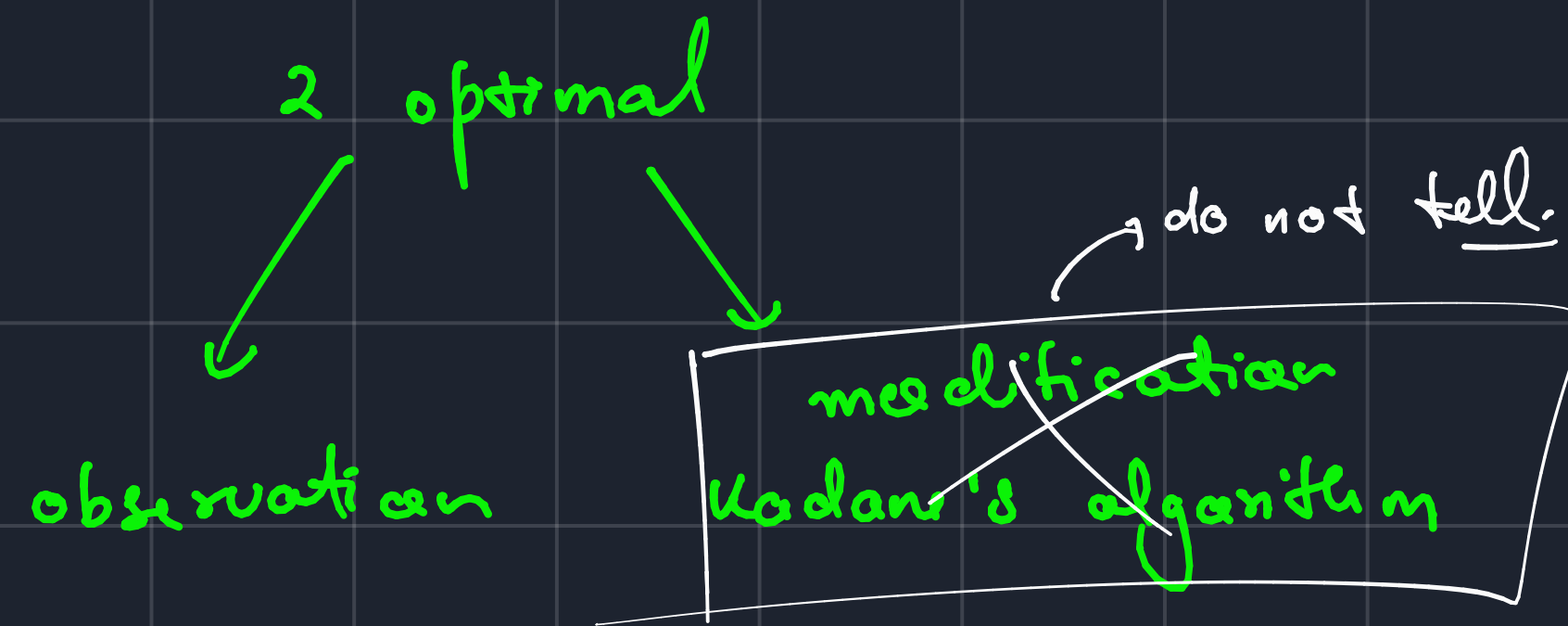
{

prod = prod * arr[j]

maxi = max(maxi, prod)

}

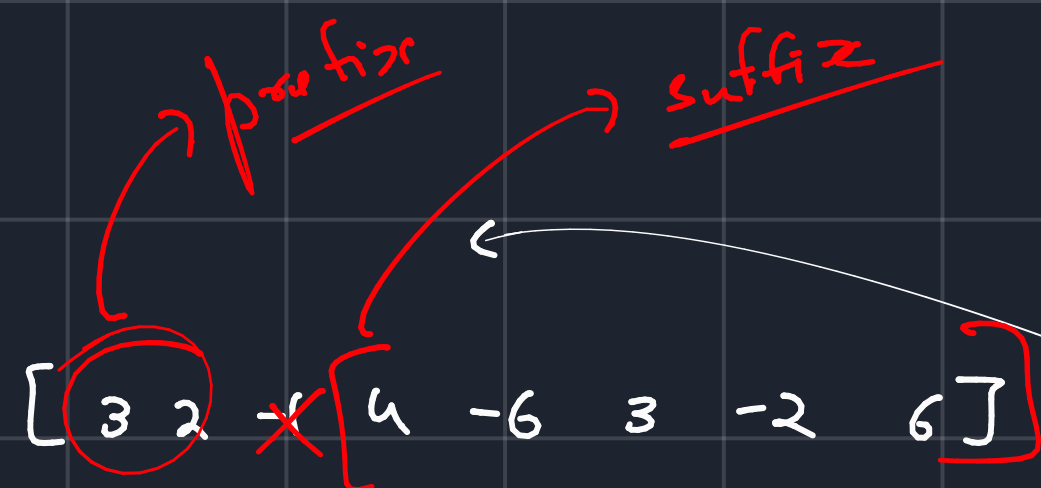
}



$$arr = [2, 3, -2, 4]$$

$$ans = 6$$

↳ observations



↳ ignoring a negative

Is an odd → the product

is positive.

→ remove the largest negative. i.e. (-1)

1) all +ve → multiply everyone

2) even negatives rest positive →

↳ multiply everyone.

3) odd negatives, rest positive.

4) what if it has zeros.

$$\text{max} = \text{INT} - \text{min}$$

$\begin{array}{cccc} 2 & 2 & 2 & 2 \\ \uparrow & \uparrow & \uparrow & \uparrow \\ 2 & 3 & -2 & 4 \end{array}$
 tried out all prer
 $\text{prer} = 6 - 12 - 48$

$\text{suff} = 1 - 8 - 24 - 48$ tried out all suff

on encountering 0 $\rightarrow$ turn it to 1.

$\text{pre} = 1, \text{suff} = 1$

for ($i = 0 \rightarrow n-1$)

$\{$

$\text{max} = \text{max}(\text{max}, \text{max}(\text{pre}, \text{suff}));$

$\text{if} (\text{pre} == 0) \text{pre} = 1;$

$\text{if} (\text{suff} == 0) \text{suff} = 1;$

$\text{prer} = \text{prer} \times \text{arr}[i]$

$\text{suff} = \text{suff} \times \text{arr}[n-i-1]$

$\}$